

time evolution

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Why dynamic formulations?

- **Static OT:** works with couplings $\pi(x, y)$
- **Dynamic OT:** studies **time evolution** of distributions
- **Motivations:**
 - Physics – mass flow, fluid mechanics
 - Probability – diffusion, stochastic control
 - ML – trajectory alignment, generative models

Static vs Dynamic OT

Static (Kantorovich)

Dynamic